

KOLMOGOROV-ARNOLD NETWORKS

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ABSTRACT

We look at Kolmogorov-Arnold Networks (KANs) and introduce some simplifications and combine different works in a novel way to improve their performance on MNIST and CIFAR10.

1. PROBLEM DEFINITION AND MOTIVATION

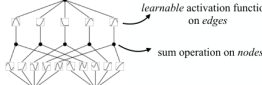
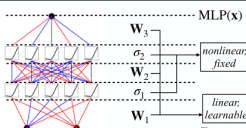
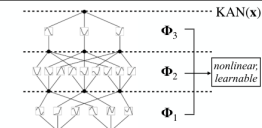
Model	Multi-Layer Perceptron (MLP)	Kolmogorov-Arnold Network (KAN)
Theorem	Universal Approximation Theorem	Kolmogorov-Arnold Representation Theorem
Formula (Shallow)	$f(\mathbf{x}) \approx \sum_{i=1}^{N(\epsilon)} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i)$	$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$
Model (Shallow)	(a) 	(b) 
Formula (Deep)	$\text{MLP}(\mathbf{x}) = (\mathbf{W}_3 \circ \sigma_2 \circ \mathbf{W}_2 \circ \sigma_1 \circ \mathbf{W}_1)(\mathbf{x})$	$\text{KAN}(\mathbf{x}) = (\Phi_3 \circ \Phi_2 \circ \Phi_1)(\mathbf{x})$
Model (Deep)	(c) 	(d) 

Fig. 1. MLP vs KAN

Kolmogorov-Arnold Networks (KANs) [1] are a new kind of neural network architecture based on the Kolmogorov-Arnold Representation Theorem. We have learnable non-linear activation functions at the edges of each layer, whose outputs are summed over at the nodes. They work comparably to standard Multi-Layer Perceptrons (MLPs), and there is a lot of room for improvement in various settings.

2. PRIOR WORK

Architecture	Details
KAN [1]	Standard architecture, non-linear functions parametrized as B-splines
MultKAN [2]	Standard KAN sums function outputs. MultKAN introduces multiplication nodes
Convolutional KAN [3]	Convolution applied to standard KAN
SKAN [4, 5]	Single-parameter non-linear functions, making it far simpler and faster

3. PROGRESS

We implement the multiplication nodes presented in MultKAN [2] to single-parameter KANs [4, 5]. The single-parameter framework of SKAN makes the training much faster at the cost of some expressivity which fully parametrized B-splines offered, and we attempt to make up for some of the lost expressivity by also adding multiplication nodes, into a MultSKAN framework. We also attempt to add the benefits offered by convolutions for image data to this framework, creating the ConvMultSKAN framework.

4. EXPERIMENTS AND RESULTS

We benchmark on MNIST and CIFAR10 with test accuracy as the metric.

Model	MNIST Test Accuracy	CIFAR-10 Test Accuracy
Simple 3-layer MLP	97.73%	53.33%
Simple CNN	99.03%	72.55%
KAN	97.31%	55.78%
SKAN	97.43%	49.61%
MultSKAN	97.71%	50.89%
ConvMultSKAN	99.21%	66.94%

5. SUMMARY OF CONTRIBUTIONS AND NOVELTY

The main contributions involve combining multiple prior works in a novel way to create the MultSKAN and ConvMultSKAN frameworks. We release the code at <https://github.com/TheReconPilot/kan-experiments>

6. RESPONSIBILITY OF INDIVIDUAL MEMBERS

All of the work was carried out by the single author of this report, Purva Parmar.

7. REFERENCES

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- [3] Alexander Dylan Bodner, Antonio Santiago Tepsich, Jack Natan Spolski, and Santiago Pourteau, “Convolutional kolmogorov-arnold networks,” 2025.
- [4] Zhijie Chen and Xinglin Zhang, “Lss-skan: Efficient kolmogorov-arnold networks based on single-parameterized function,” 2024.
- [5] Zhijie Chen and Xinglin Zhang, “Larctan-skan: Simple and efficient single-parameterized kolmogorov-arnold networks using learnable trigonometric function,” 2024.